Automate your GIS – Scripting in Python
5 credits (hp)

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General info

Part of the course will be together with the master's course NGN220 – Open-Source GIS

We will use GitHub for distributing data, code and other information.

https://github.com/biglimp/PhDCourseHT26

Course outline

Course focusing on hands-on GIS

1. First week – Practical tutorials and exercises
2. Second and third week – Project work

Python programming
Python and GIS
Programming and scripting using QGIS
QGIS (processing) plugin building
Tools for spatial manipulation/processing
ArcGIS and Python

Project work

Preferable something related to your PhD-work

For example:

- Big data handling
- Automate GIS processing/analysis
- Build a plugin for your spatial model
- Tryout and demonstrate certain GIS-technique using command line and scripting
- Make a tutorial based on some spatial library available

Hand in the form of a written report including your code and results

Presentation of your work in the final seminar

Personal Computer configuration

We will make use of OSGeo packages which includes QGIS, GRASS, SAGA and other software.

We will mainly use Visual Studio Code for coding.

OSGeo is delivered with its own Python environment.

Instructions on how to install and configure your computer can be found in our GitHub repository.

Before your first lecture we can assist you in configuring (check) your computer.

GOOD LUCK and HAVE FOR CODING!